

Symmetry-Aware Raman and Infrared Spectroscopy in TD-SCHA

API unification, group-theoretical reduction, validation, and audit

TD-SCHA development report

5 August 2026

Abstract

This report documents a new, backend-neutral optical-spectroscopy layer for TD-SCHA. It replaces a confusing collection of Raman and infrared (IR) preparation routines with one restartable `Spectroscopy` workflow for polarized and unpolarized observables. The physical perturbations are defined once; real-space, reciprocal-space, and atom-Fourier Lanczos classes only implement representation-specific numerical hooks. A finite-group orbit analysis removes symmetry-equivalent external perturbations. Within every remaining Lanczos application, a stabilizer projector and a right-coset decomposition reduce the expensive symmetrized ensemble average without changing it.

On the bundled cubic two-atom, ten-configuration test ensemble, the controlled unpolarized Raman benchmark has 7 requested invariant components, of which 3 are nonzero and only 1 is independent. Unpolarized IR similarly falls from three runs to 1. New and legacy spectra agree to relative L^∞ errors of $3.04\text{e-}14$ (Raman) and $3.05\text{e-}14$ (IR). For 256 Lanczos steps, measured end-to-end speedups are $9.88\times$ and $9.87\times$, respectively. At 640 configurations, the isolated anharmonic kernels agree with the full group below 10^{-13} relative error and are $12.70\times$ faster in real space and $11.40\times$ faster in q space. A configuration-scaling study explains why the same ratios are only about 1.9 and 1.6 at ten configurations.

The audit also removes the obsolete kernel-polynomial-method (KPM) backend, quarantines unvalidated two-phonon optical code behind `NotImplementedError`, and fixes zero-valued, symmetry-forbidden Raman channels so they contribute exactly zero without launching an invalid Lanczos recursion. This is a stand-alone spectroscopy report: it is separate from, and does not replace, the interpolation report.

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1 Scope and design outcome

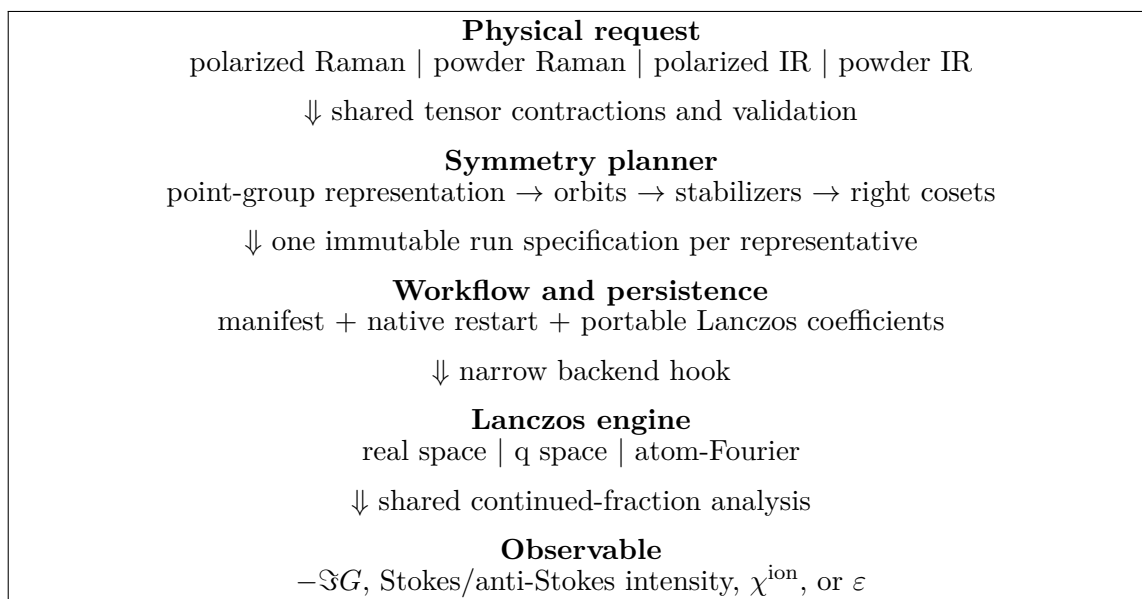
The stable user-facing object is `tdscha.Spectroscopy.Spectroscopy`. It owns four tasks that were previously mixed into mutable Lanczos engines:

1. defining the requested Raman or IR observable;
2. finding symmetry-equivalent perturbations and independent runs;
3. executing and restarting those runs; and
4. reconstructing response, Stokes/anti-Stokes Raman spectra, and the IR dielectric function.

Lanczos classes remain numerical engines. This separation mirrors the role of the Hessian driver: the high-level class expresses the physical task while the backend performs applications of the TD-SCHA linear operator.

The implementation supports three backends: `real`, `qspace`, and `atom_fourier`. KPM is not a fourth option: its source, build entry, CLI handling, tests, and documentation were removed because it was unmaintained and no longer useful for this code base. The two-phonon Raman source survives only as an inert private archival file, not as a callable implementation.

1.1 A compact view of the architecture



No backend duplicates the Placzek components, Born-charge contraction, request planning, restart protocol, or spectrum assembly. Conversely, the workflow does not duplicate Lanczos recurrence or continued-fraction code.

2 Audit of the legacy Raman API

2.1 Why the old API was confusing

The legacy class exposed two similarly named methods which did not compute a complete unpolarized spectrum:

- `prepare_raman()` defaults to a *polarized* xx perturbation. Its optional `unpolarized=i` selector prepares one of seven *normalized* invariant components.
- `prepare_unpolarized_raman(index=0)` also prepares only one component, but in a *raw Cartesian* normalization. Despite its name, one call never computes the powder spectrum.

Both variants are mathematically correct in the present branch only when paired with their matching weights. They are summarized in Table 1.

Table 1: Equivalent component conventions. R_{ab} denotes a Raman derivative vector, not a scalar mode intensity.

j	component	normalized vector; weight	raw vector; weight
0	isotropic	$(R_{xx} + R_{yy} + R_{zz})/3$; 45	$R_{xx} + R_{yy} + R_{zz}$; 5
1	diagonal	$(R_{xx} - R_{yy})/\sqrt{2}$; 7	$R_{xx} - R_{yy}$; 7/2
2	diagonal	$(R_{xx} - R_{zz})/\sqrt{2}$; 7	$R_{xx} - R_{zz}$; 7/2
3	diagonal	$(R_{yy} - R_{zz})/\sqrt{2}$; 7	$R_{yy} - R_{zz}$; 7/2
4–6	off-diagonal	$\sqrt{3}R_{xy,xz,yz}$; 7	$R_{xy,xz,yz}$; 21

For each row, scaling the perturbation by s scales its diagonal response by s^2 . Therefore $45(1/3)^2 = 5$, $7(1/\sqrt{2})^2 = 7/2$, and $7(\sqrt{3})^2 = 21$. The two columns produce exactly the same total if, and only if, the corresponding weights are used. Combining normalized vectors with raw weights (or the reverse) is wrong.

The old `mixed=True` option adds two polarization contractions *coherently* before taking the response. This is not the same as adding two spectra. The stable API removes that ambiguity: a general coherent contraction is a single symmetric 3×3 Raman coefficient tensor, while separate measurements are separate named requests.

2.2 Was one implementation wrong?

Historically, yes. Before the 1.7 hotfix, the normalized `prepare_raman(unpolarized=i)` path accumulated channels 0–3 and then overwrote the accumulated vector with the final Cartesian term. In particular, channels 2 and 3 became identical and R_{xx} disappeared from the powder result. Saved normalized channels 0–3 from that version cannot be corrected after the fact because the missing diagonal cross terms were never computed; they must be rerun. Channels 4–6 were unaffected.

The present implementation has one shared vector builder and characterization tests for both conventions. Thus the current answer is:

The normalized and raw one-phonon methods are both correct, but they are two normalizations of the same seven-term decomposition. Their former duplicate implementations and names made correct use unnecessarily difficult, and one duplicate did contain a serious overwrite bug.

3 Linear-response quantities

3.1 Coordinates, perturbations, and the Green function

Consider a primitive cell with N atoms. Let $\mathbf{u} \in \mathbb{R}^{3N}$ collect Cartesian displacements and let

$$\mathbf{M} = \text{diag}(M_1, M_1, M_1, \dots, M_N, M_N, M_N) \quad (1)$$

be the mass matrix. An external generalized force has the linear coupling

$$\delta \hat{H}(t) = -f(t) \mathbf{v}^\top \mathbf{u}, \quad (2)$$

where $\mathbf{v}$ is the Cartesian optical vertex. The backend maps this vertex to its mass-weighted normal-coordinate representation. This map, including the factor associated with a repeated supercell-periodic Γ pattern, is backend-specific; the definition of $\mathbf{v}$ is not.

In the Wigner TD-SCHA formulation [2], the linearized dynamics is represented by a linear operator $\mathcal{L}_W$ acting on the one- and two-phonon response variables. For a prepared Krylov vector $|p\rangle$, the code evaluates a projected resolvent of the form

$$G_p(\omega) = \langle p | [\mathcal{L}_W + (\omega + i\eta)^2]^{-1} | p \rangle, \quad (3)$$

with the exact internal sign transformation documented by the Wigner Lanczos implementation. The nonnegative, Bose-free spectral response used by the public API is

$$\mathcal{R}_p(\omega) = -\Im G_p(\omega), \quad \omega > 0. \quad (4)$$

Lanczos tridiagonalization [3] turns Equation (3) into a continued fraction. The new workflow calls that existing implementation; it does not introduce a second spectral evaluator.

3.2 Equilibrium one-phonon scope

This report concerns an equilibrium optical vertex that is linear in displacement. Schematically,

$$A(\mathbf{u}) = A(\mathbf{0}) + \mathbf{v}^\top \mathbf{u} + O(u^2). \quad (5)$$

Configuration-dependent effective charges and second derivatives of the polarizability would produce explicit two-phonon optical vertices. The legacy implementation of those terms did not have a sufficiently audited observable definition, units, or symmetry action. It is therefore disabled rather than silently mixed with the validated one-phonon response.

4 Raman observables

4.1 Polarized Raman vertex

Let α_{ab} be the electronic polarizability tensor and define its equilibrium derivative

$$P_{ab,i} = \left. \frac{\partial \alpha_{ab}}{\partial u_i} \right|_{\mathbf{u}=0}, \quad a, b \in \{x, y, z\}, \quad i = 1, \dots, 3N. \quad (6)$$

For incident and analyzed unit polarization vectors $\mathbf{e}_{\text{in}}$ and $\mathbf{e}_{\text{out}}$, nonresonant Raman scattering probes the symmetric coefficient matrix

$$\mathbf{C} = \frac{1}{2} (\mathbf{e}_{\text{in}} \mathbf{e}_{\text{out}}^\top + \mathbf{e}_{\text{out}} \mathbf{e}_{\text{in}}^\top). \quad (7)$$

The prepared Cartesian vertex is the linear contraction

$$v_i(\mathbf{C}) = \sum_{ab} C_{ab} P_{ab,i}. \quad (8)$$

This equation is the single source of truth used by `add_raman_polarized` and `add_raman_tensor`. Antisymmetric/resonant Raman tensors are outside this API because their rotational invariants differ.

4.2 Unpolarized Placzek invariant

For a symmetric Raman tensor R associated with one response channel, define

$$\bar{\alpha} = \frac{R_{xx} + R_{yy} + R_{zz}}{3}, \quad (9)$$

$$\gamma^2 = \frac{1}{2} [(R_{xx} - R_{yy})^2 + (R_{xx} - R_{zz})^2 + (R_{yy} - R_{zz})^2] + 3(R_{xy}^2 + R_{xz}^2 + R_{yz}^2). \quad (10)$$

The conventional rotationally averaged nonresonant intensity is

$$I_{\text{powder}} = 45 \bar{\alpha}^2 + 7 \gamma^2. \quad (11)$$

The same invariant is used in first-principles powder Raman methods; polar materials may require additional nonanalytic/electro-optic physics beyond this basic average [5].

Equation (11) is a sum of seven diagonal response functions, not seven arbitrary light polarizations. Define the coefficient tensors $C^{(j)}$ by the normalized column of Table 1. Then

$$\mathcal{R}_{\text{powder}}(\omega) = 45 \mathcal{R}_{v(C^{(0)})}(\omega) + 7 \sum_{j=1}^6 \mathcal{R}_{v(C^{(j)})}(\omega). \quad (12)$$

The API records every term, including an exactly zero $v(C^{(j)})$. A zero term contributes zero and creates no numerical run.

4.3 Response, Stokes, and anti-Stokes output

With

$$n_B(\omega, T) = \frac{1}{\exp(\hbar\omega/k_B T) - 1}, \quad (13)$$

the thermal factors returned by the implementation are

$$I_S(\omega) = [n_B(\omega, T) + 1] \mathcal{R}(\omega), \quad (14)$$

$$I_{AS}(\omega) = n_B(\omega, T) \mathcal{R}(\omega). \quad (15)$$

Consequently $I_S - I_{AS} = \mathcal{R}$ and $I_{AS}/I_S = \exp(-\hbar\omega/k_B T)$. If a laser frequency ω_L is supplied in the same units as ω , the API additionally multiplies by $(\omega_L - \omega)^4$ for Stokes or $(\omega_L + \omega)^4$ for anti-Stokes. Absolute cross sections still require the experiment-dependent prefactor and a consistent polarizability unit.

5 Infrared observables

5.1 Born effective-charge vertex

The Born effective-charge tensor connects a displacement of atom κ to the macroscopic polarization:

$$Z_{\kappa, ai}^* = \Omega \left. \frac{\partial P_a}{\partial u_{\kappa i}} \right|_{\mathcal{E}=0}, \quad (16)$$

in the atomic-unit convention used by CellConstructor. For a unit electric field direction $\mathbf{e}$, the Cartesian IR vertex is

$$v_{\kappa i}(\mathbf{e}) = \sum_a e_a Z_{\kappa, ai}^*. \quad (17)$$

The powder response is the Cartesian trace average

$$G_{\text{powder}}(\omega) = \frac{1}{3} \sum_{a=x,y,z} G_{v(\mathbf{e}_a)}(\omega). \quad (18)$$

In cubic symmetry the three terms belong to one orbit, so one Lanczos run is sufficient.

5.2 Ionic susceptibility and electronic background

The raw Lanczos Green function uses a Rydberg energy convention. Because $1 \text{ Ha} = 2 \text{ Ry}$, the projected ionic susceptibility exposed by the API is

$$\chi_e^{\text{ion}}(\omega) = 2G_{v(e)}(\omega) \quad (19)$$

in Hartree atomic units. Let ϵ_∞ be the clamped-ion electronic dielectric tensor. The projected total dielectric response is

$$\epsilon_e(\omega) = \mathbf{e}^\top \epsilon_\infty \mathbf{e} + \frac{4\pi}{\Omega_{\text{bohr}^3}} \chi_e^{\text{ion}}(\omega). \quad (20)$$

For the unpolarized request, the electronic term is $\text{Tr}(\epsilon_\infty)/3$. Cell volume is read in $\text{\AA}^3$ and converted to bohr^3 before applying Equation (20). An explicit `ionic_prefactor` override is available for a different electromagnetic convention; no hidden material-dependent scaling is applied.

The retarded response convention in the present evaluator is reported through $-\Im G \geq 0$. If an absorption coefficient or optical conductivity is formed downstream, the sign must be kept consistent with that Fourier convention. The stable layer returns the complex dielectric function rather than guessing SI/cgs conversion, refractive-index branch, sample geometry, or experimental broadening.

6 Finite groups as matrices

This section develops the symmetry reduction using only finite-dimensional linear algebra. No prior abstract-algebra course is assumed.

6.1 Group and representation

A finite symmetry group $G = \{g_1, \dots, g_{|G|}\}$ is a set of operations closed under composition, with an identity and inverses. To act on a Cartesian displacement vector, each operation is represented by an orthogonal matrix

$$D(g) \in \mathbb{R}^{3N \times 3N}, \quad D(g_1 g_2) = D(g_1) D(g_2), \quad D(g)^\top D(g) = I. \quad (21)$$

The matrix combines a 3×3 Cartesian rotation with a permutation of symmetry-equivalent atoms. The implementation obtains crystallographic operations with `spglib` [6], filters out rotations incompatible with the finite supercell mesh, and then constructs and verifies Equation (21) numerically.

For example, a 90° rotation about z acts on a vector by

$$R_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_z \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (22)$$

For many atoms, copies of R_z appear in a larger matrix and are connected by the atom permutation.

6.2 Orbits remove duplicate external calculations

Given a nonzero prepared perturbation $\mathbf{v}$, its sign-aware orbit is

$$\mathcal{O}_\mathbf{v} = \{ \chi D(g) \mathbf{v} : g \in G, \chi \in \{+1, -1\} \}. \quad (23)$$

The sign is harmless for a diagonal response because $G_{-v, -v} = G_{v, v}$, but it is stored for future cross-response work. Two requested Cartesian vertices share one Lanczos run only if their complete vectors have equal norm and match under an actual $D(g)$ to tolerance. The planner never assumes that labels such as “ x ” and “ y ” are equivalent.

For cubic IR,

$$\{\mathbf{v}_x, \mathbf{v}_y, \mathbf{v}_z\} = \mathcal{O}_{\mathbf{v}_x} \quad (24)$$

up to signs; thus three requested terms need one representative. For a generic low-symmetry crystal the three vectors need not share an orbit and no such reduction is made.

6.3 Stabilizer and orbit–stabilizer counting

The sign-aware stabilizer (little group) of $\mathbf{v}$ is

$$H_{\mathbf{v}} = \{h \in G : D(h)\mathbf{v} = \chi(h)\mathbf{v}, \quad |\chi(h)| = 1\}. \quad (25)$$

It is a subgroup. At Γ the optical perturbations are real and $\chi(h) = \pm 1$. At finite non-time-reversal-invariant momentum the Bloch representation is complex and $\chi(h)$ can be any unit phase. In either case it is a one-dimensional representation because

$$\chi(h_1 h_2) = \chi(h_1) \chi(h_2). \quad (26)$$

The orbit–stabilizer theorem becomes a simple counting identity:

$$|\mathcal{O}_{\mathbf{v}}| = \frac{|G|}{|H_{\mathbf{v}}|}. \quad (27)$$

In the cubic benchmark, $|G| = 48$ and $|H_{\mathbf{v}}| = 16$, so the orbit contains $48/16 = 3$ Cartesian directions.

6.4 Right cosets reduce the inner ensemble average

Choose one representative r_c from each right coset

$$H_{\mathbf{v}} r_c = \{h r_c : h \in H_{\mathbf{v}}\}, \quad G = \bigsqcup_{c=1}^m H_{\mathbf{v}} r_c, \quad m = |G|/|H_{\mathbf{v}}|. \quad (28)$$

The disjoint-union symbol means every $g \in G$ appears in exactly one coset. Define the character-weighted projector

$$P_H^{(\chi)} = \frac{1}{|H|} \sum_{h \in H} \chi(h)^* D(h). \quad (29)$$

From the representation rule in Equation (21), one verifies

$$\left(P_H^{(\chi)}\right)^2 = P_H^{(\chi)}, \quad D(h)P_H^{(\chi)} = \chi(h)P_H^{(\chi)}. \quad (30)$$

Thus $P_H^{(\chi)}$ selects the part transforming like the perturbation.

Let $\mathbf{f}_{r_c}$ and K_{r_c} be, respectively, the perturbed force vector and second-derivative matrix obtained after applying only coset representative r_c to an ensemble configuration. Equivariance rewrites the full group average as

$$\bar{\mathbf{f}} = \frac{1}{m} \sum_{c=1}^m \frac{1}{|H|} \sum_{h \in H} \chi(h)^* D(h) \mathbf{f}_{r_c}, \quad (31)$$

$$\bar{K} = \frac{1}{m} \sum_{c=1}^m \frac{1}{|H|} \sum_{h \in H} \chi(h)^* D(h) K_{r_c} D(h)^{\top}. \quad (32)$$

Equations (31)–(32) are not an approximation. They rearrange the same $|G|$ terms into m expensive ensemble transformations followed by a cheap matrix projector. Right rather than left cosets are required by the order in which this implementation acts on the vectors; the multiplication table and the matrix representation are tested together to prevent an ordering mistake.

7 Implementation details

7.1 One source of truth for perturbations

`Modules/Spectroscopy.py` defines immutable perturbation objects and the seven Raman components. The contractions in Equations (8) and (17) occur once. Legacy methods delegate to those functions and then call a single backend hook, `_prepare_gamma_cartesian_perturbation`.

Real space tiles the unit-cell vertex over the supercell and performs the existing mass/normal-mode projection. Q space inserts the vertex at Γ , with its established $\sqrt{N_{\text{cell}}}$ normalization. `QSpaceLanczos` does not redeclare public Raman or IR methods. `QSpaceAtomFourierLanczos` inherits the q-space optical preparation and overrides only the fine/coarse numerical representation needed by its operator. This is deliberate: atom-Fourier interpolation is a backend, not a second spectroscopy API.

7.2 Planning and global deduplication

For every request the planner:

1. contracts the optical tensor into a complete Cartesian vector;
2. records exactly zero vectors as zero contributions;
3. partitions nonzero vectors into sign-aware orbits;
4. creates one stable content-derived run identifier per representative;
5. stores stabilizer indices, characters, right cosets, assembly weights, and original component indices; and
6. deduplicates representatives across different named requests.

The zero-vector step was added during this audit. Rejecting all zero vectors is sensible for a low-level Lanczos engine, because their norm cannot normalize a Krylov recurrence. It was wrong at the high level: a symmetry-forbidden optical tensor component is a valid term whose response is identically zero.

7.3 Kernel integration

The real-space and q-space Julia kernels accept three optional arrays: selected coset symmetry indices, stabilizer indices, and stabilizer characters. Empty arrays select the legacy full average. Otherwise the kernel evaluates only the selected ensemble replicas and applies Equations (31) and (32) to both the force and every second-derivative block.

The q-space implementation reconstructs the complete sparse block operator, applies $D(h)KD(h)^T$, and extracts the independent momentum-pair blocks again. The atom-Fourier backend calls the same q-space projector after its fine/coarse mapping. Unsupported execution modes or a nonreducing stabilizer automatically use the full average.

For optical Γ requests, the workflow calls `configure_spectroscopy_symmetry`. For a manually prepared finite-q perturbation, `QSpaceLanczos.configure_qspace_perturbation_symmetry` instead applies each exact cached sparse Bloch matrix to the one-phonon input, detects its invariant complex line, and constructs the little group, right cosets, and conjugated unit-phase projector coefficients. An element mapping q outside its star component cannot belong to this stabilizer. Because an inner-group operation preserves total momentum, its rank-two action maps the stored pairs $q_1 + q_2 = q$ back into the same bilinear pair sector; transpose, not Hermitian conjugation, is therefore the correct second-index action.

The public `Spectroscopy` constructor also promotes `ignore_v3`, `ignore_v4`, and `lo_to_split` out of an opaque backend dictionary. The old dictionary spelling is still parsed for script compatibility.

Atom-Fourier now passes the chosen nonanalytic Γ direction through the parent q-space basis, pins commensurate fine points to that basis, and keeps the tensorial dipole–dipole term between coarse points. With `ignore_effective_charges=True`, this harmonic backend instead omits the entire dipolar interpolation, including the directional Γ limit. This uses a private dynamical-matrix copy: the original Z^* remains available for the IR vertex.

7.4 Restart and load-only analysis

Each independent run has

```
spectroscopy/
  manifest.json
  runs/<stable-id>/
    status.npz
    result.npz
    lanczos.abc
    metadata.json
```

The manifest includes the request definitions, backend and options, structure, temperature, optical tensors through an ensemble fingerprint, orbit and coset metadata, completed steps, convergence state, and errors. JSON, native status, and portable coefficient files are written to temporary siblings and committed with an atomic rename. Only the MPI master writes; barriers keep ranks at the same state transition.

`Spectroscopy.load` reconstructs all named observables from completed representatives. A portable `.abc` file is sufficient for load-only analysis because its perturbation modulus and the sign/shift/Wigner conventions are recorded. Resuming the recurrence uses native status and a compatible ensemble. A changed request, backend option, run option, or ensemble fingerprint is rejected rather than guessed.

8 Stable user API

```
from tdscha.Spectroscopy import Spectroscopy

job = Spectroscopy(
    ensemble,
    backend="qspace", # "real", "qspace", "atom_fourier"
    workdir="optical_spectroscopy",
    use_symmetries=True, # the default
)

job.add_raman_polarized([1, 0, 0], [0, 1, 0], name="raman_xy")
job.add_raman_unpolarized(name="raman_powder")
job.add_ir_polarized([1, 0, 0], name="ir_x")
job.add_ir_unpolarized(name="ir_powder")

print(job.plan_calculations())
job.run(500, save_each=10, resume=True)
```

Analysis can occur in the same process or from disk:

```
import numpy as np
from tdscha.Spectroscopy import Spectroscopy

loaded = Spectroscopy.load("optical_spectroscopy")
omega = np.linspace(1e-5, 0.01, 2000) # Rydberg frequency units
opts = dict(smearing=2e-5, use_terminator=False)
```

```

response = loaded.raman_spectrum(
    "raman_powder", omega, kind="response", **opts)
stokes = loaded.raman_spectrum(
    "raman_powder", omega, kind="stokes", **opts)
anti = loaded.raman_spectrum(
    "raman_powder", omega, kind="anti_stokes", **opts)

epsilon = loaded.dielectric_function(
    "ir_powder", omega, **opts) # epsilon_inf inferred from the dyn/manifest

```

For a custom coherent Raman contraction use `add_raman_tensor(C,name)`. For an already contracted vertex use `add_raman_vector` or `add_ir_vector`; an explicit IR vector requires an explicit electronic projection when forming ε , because its field direction is no longer inferable. The complete 3×3 electronic tensor is retained: a polarized result uses $\mathbf{e}^T \varepsilon_\infty \mathbf{e}$, while a powder result uses $\text{Tr} \varepsilon_\infty / 3$. Passing `epsilon_infinity=` remains a tensor override.

9 Numerical validation

9.1 Test construction

The reproducible driver is `spectroscopy_report/benchmark_spectroscopy.py`. It loads the repository’s cubic SnTe ensemble at 250 K: two atoms in the primitive cell, a $2 \times 2 \times 2$ supercell, and ten stochastic configurations. To test the API without claiming a material-specific optical prediction, controlled vertices are attached:

- isotropic neutral effective charges $Z_1^* = I$, $Z_2^* = -I$;
- a synthetic Raman derivative for which trace and the three diagonal deviatoric channels vanish, while xy , xz , and yz prepare three symmetry-equivalent opposite-sublattice displacements.

This is a software-verification case, not a predicted Raman spectrum of SnTe. The same ensemble, operator, 256 recurrence steps, frequency grid, and smearing are used on both sides.

The legacy Raman reference invokes the direct low-level API separately for each of its 3 nonzero channels and manually combines them with the normalized weights. Its four zero channels are skipped manually because the low-level API cannot normalize a zero Krylov vector. The new calculation receives all 7 components, records four zero contributions, and schedules 1 run. The legacy IR reference schedules x, y, z independently; the new request schedules 1 run.

9.2 Spectral agreement

The agreement in Figures 1 and 2 is much tighter than any physical statistical error in an SSCHA ensemble. More importantly, the test suite compares a single reduced application of the anharmonic operator with the complete group average before Lanczos spectral analysis. The real-space relative error is $8.47\text{e-}14$ and the q-space error is $2.70\text{e-}14$. This rules out the possibility that two visibly broadened spectra merely hide different kernels.

9.3 Additional correctness gates

The focused suite exercises:

- exact normalized/raw Raman invariant equivalence;
- polarized and coherent tensor contractions;
- real/q-space vertex and perturbation-modulus parity;

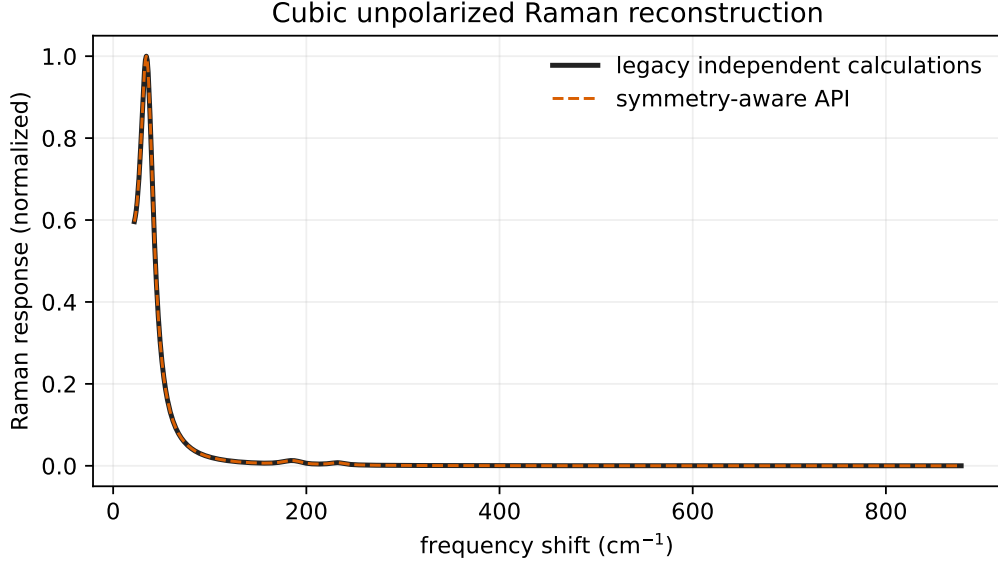


Figure 1: Controlled cubic unpolarized Raman response. The legacy sum uses three independent nonzero calculations; the new result reconstructs all seven requested terms from one calculation and four exact zeros. Curves are visually coincident; the relative L^∞ discrepancy is $3.04\text{e-}14$.

- cubic orbit reduction and anisotropic-mesh filtering;
- real/q-space full-average versus stabilizer/coset parity;
- direct q-space full-replica versus coset/inner-group parity on every point of a cubic $3 \times 3 \times 3$ mesh: Γ and 26 non-TRI points, followed by a second application with a generated nonzero two-phonon input sector;
- atom-Fourier execution through inherited q-space preparation;
- directional atom-Fourier LO-TO basis pinning and interpolation-local effective-charge suppression that retains Z^* for IR preparation;
- interrupted native restart, strict manifest mismatch rejection, portable `.abc` load, and load-only analysis;
- Bose detailed balance, Stokes minus anti-Stokes identity, and anisotropic tensorial IR electronic-background handling, including load-only inference and explicit override;
- zero symmetry-forbidden Raman channels; and
- uniform failure of every quarantined two-phonon/configuration- dependent optical entry point in base and derived classes.

At report generation time the focused spectroscopy/Raman/IR/interpolation suite contains 69 passing tests. The complete non-heavy repository result is recorded in the project handoff after the final audit run, so this document does not misstate a count that changes when obsolete KPM tests are deleted.

10 Performance analysis

Three effects contribute to the workflow improvement:

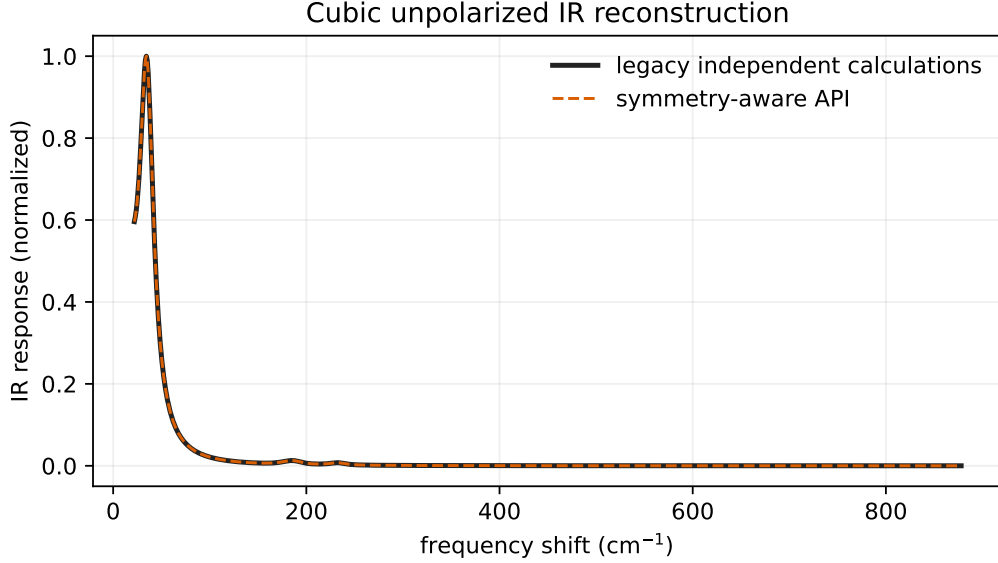


Figure 2: Controlled cubic unpolarized IR response. Three legacy Cartesian calculations are reconstructed from one symmetry representative. The relative L^∞ discrepancy is $3.05\text{e-}14$.

Table 2: End-to-end benchmark, including conservative new-workflow manifest and checkpoint writes. Times are wall seconds on the report-generation host, not portable hardware constants.

observable	legacy	new	speedup
unpolarized Raman	14.645	1.482	$9.88\times$
unpolarized IR	14.580	1.477	$9.87\times$

1. **Across observables:** three equivalent Raman or IR channels are represented by one run.
2. **Translations:** a Γ optical perturbation separates the cheap translation projector from the expensive point-group ensemble replication. The direct legacy real-space path used the full supercell space-group list.
3. **Within one run:** 48 point-group operations reduce to 3 expensive representatives, followed by the 16-term linear-algebra projector.

These factors are not expected to multiply into an exact wall-time ratio: initialization, Python/Julia calls, BLAS work, checkpoint I/O, and the stabilizer projector remain. In particular, the projector is independent of the number of configurations and performs 16 sparse vector/rank-two transformations after the representative average. With only ten configurations that fixed work is comparable to evaluating three representatives, giving the initially puzzling $1.87\times$ (real) and $1.58\times$ (q-space) speedups. At 40, 160, and 640 configurations the measured real-space ratios are 3.86, 8.83, and 12.70; q-space gives 3.23, 7.16, and 11.40. This monotone approach toward $48/3 = 16$ verifies that the optimization scales as predicted rather than hiding a replicated-ensemble calculation in the projector. The isolated-kernel speedups in Table 3 are therefore the clean measure of the new coset optimization, while Table 2 measures what a legacy script and a restartable production workflow actually experienced on this fixture.

A six-step microbenchmark was also inspected during the audit. There, atomic checkpoint and manifest writes dominated and made the new workflow slower than three unsaved legacy calls even though its kernel was faster. This is expected fixed-cost behavior, not a hidden regression. The reported 256-step case is still small but makes recurrence work visible.

Table 3: Isolated anharmonic-operator benchmark at 640 configurations. Medians use seven repeated applications after warm-up.

backend	full $ G $	representatives	stabilizer $ H $	speedup
real space	48	3	16	$12.70\times$
q space	48	3	16	$11.40\times$

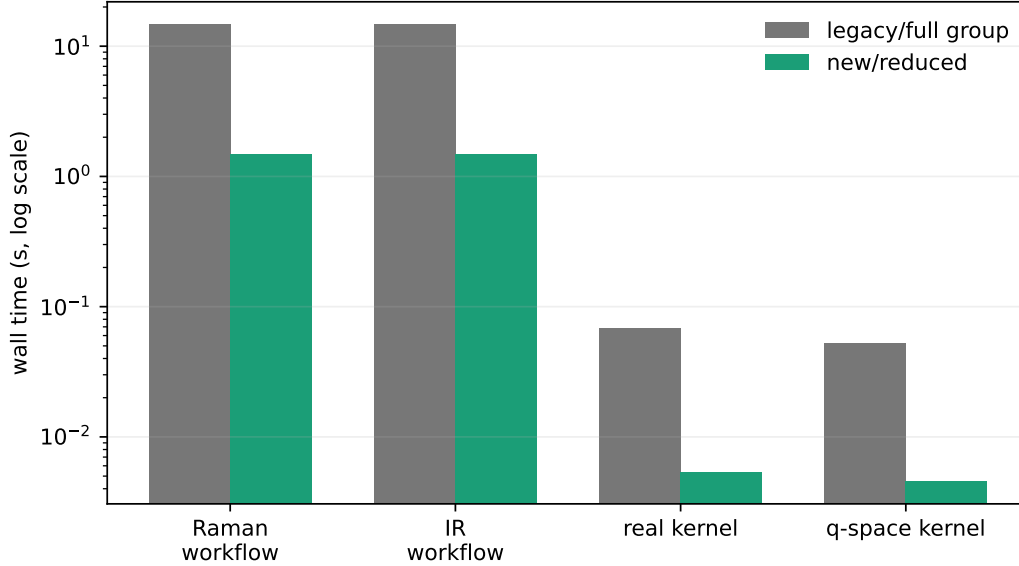


Figure 3: Measured workflow and isolated-kernel wall times. The logarithmic axis is necessary because a complete 256-step workflow and one kernel application have different natural scales.

11 Code audit and removal decisions

11.1 Findings corrected in this branch

area	audit result
Raman definitions	One canonical component table now generates both legacy normalizations and the new API. The former channel-accumulation overwrite cannot recur in a second copy.
Derived classes	QSpaceLanczos and QSpaceAtomFourier do not duplicate public Raman/IR preparation. Only representation-specific Γ preparation and operator application remain overridden.
Finite q	The q-space backend now retains its exact sparse Bloch representation and detects finite-q little groups with complex characters. Full/coset parity is checked for all 27 points of an odd cubic mesh and for a nonzero two-phonon input.
Zero channels	Exactly zero Raman derivatives are valid symmetry-forbidden contributions. The planner now records them with no run identifier, and the evaluator skips them as mathematical zeros.
Scale independence	Symmetry matching and canonical signs use relative vector tolerances. Roundoff-level forbidden components are detected relative to the strongest channel in the same request, so a physically small but nonzero optical tensor is not quantized away.

area	audit result
Global deduplication	Orbit analysis is performed over all named requests at once. Separately added cubic <code>ir_x</code> and <code>ir_y</code> requests now share one run while retaining independent reconstruction weights.
Group order	Unit-cell crystallographic rotations are matched to the backend's native ordering by matrix comparison. The code does not assume <code>spglib</code> and <code>CellConstructor</code> enumerate operations identically.
Coset side	Right cosets are used consistently with the representation action and checked through full-average parity tests.
Rank-two output	The stabilizer is applied to both force and second-derivative terms, including every q-space momentum-pair block.
Restart writes	Native status uses a temporary filename that preserves the <code>.npz</code> extension before atomic replacement; interruption/resume is tested.
IR units	The raw Rydberg Green function receives the explicit factor two for Hartree susceptibility. $4\pi/\Omega$ uses bohr ³ , and the electronic dielectric projection uses the full tensor, inferred from the dynamical matrix or stored manifest unless explicitly overridden.
Backend switches	<code>ignore_v3</code> , <code>ignore_v4</code> , and <code>lo_to_split</code> are public, manifest-recorded options; legacy <code>backend_options</code> scripts remain accepted. Atom-Fourier implements directional LO-TO interpolation rather than raising <code>NotImplementedError</code> .
KPM	Backend module, build/install entry, CLI/script loaders, documentation, and KPM tests were purged. Repository-wide textual audit finds no KPM API reference. Git history remains the recovery mechanism.
Two-phonon optics	Public legacy names remain only to fail loudly with <code>NotImplementedError</code> . The Raman source backup is private, unimported, and excluded from installation.

11.2 Deliberate limitations

The following are not silently approximated:

- configuration-dependent/two-phonon Raman and IR vertices;
- resonant or antisymmetric Raman tensors;
- cross-correlations between two distinct external vertices;
- absolute experimental Raman cross sections;
- automatic SI absorption or conductivity conversions; and
- interpreting the controlled benchmark as a material prediction.

The stable API covers equilibrium one-phonon polarized/unpolarized Raman and IR response. Adding any item above should begin from a written observable definition and symmetry transformation law, then acquire full-average parity tests before being exposed.

12 Reproducibility

From the repository root, regenerate numerical data and vector figures with

```
micromamba run -n sscha \
python spectroscopy_report/benchmark_spectroscopy.py
```

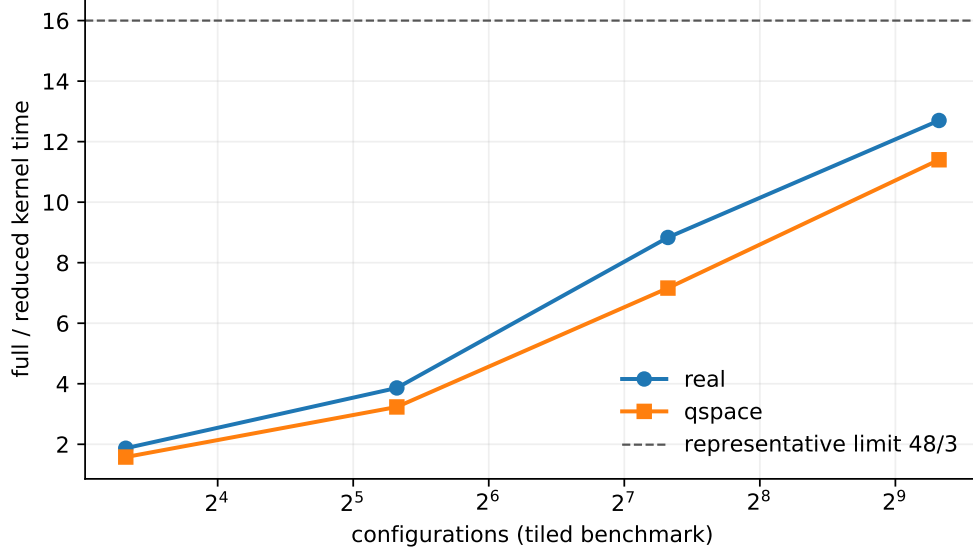


Figure 4: Measured kernel speedup after deterministically tiling the same ten configurations. Tiling leaves the normalized ensemble average unchanged and isolates computational scaling; it does not claim improved statistical sampling. The dashed line is the asymptotic representative ratio $48/3 = 16$.

The script writes `data/benchmark_summary.json`, `data/spectra.csv`, generated $\text{\LaTeX}$ macros, and the four PDF figures. Compile this report from its own directory:

```
lualatex -interaction=nonstopmode -halt-on-error \
  tdscha_spectroscopy_report.tex
lualatex -interaction=nonstopmode -halt-on-error \
  tdscha_spectroscopy_report.tex
```

This directory is intentionally independent of `report/interpolation/`. Neither its source, figures, data, nor PDF is shared with the interpolation report.

13 Conclusions

The refactor makes the default behavior unambiguous: users name a polarized or unpolarized Raman/IR observable, inspect a symmetry plan, and run it through one restartable driver. Legacy equilibrium calls continue to work, but no longer define physics independently. The two formerly competing unpolarized Raman routines are recognized as equivalent normalizations; the historically broken normalized implementation is repaired through shared construction and regression tests.

The group-theory optimization is exact linear algebra. Orbits remove equivalent requested experiments; stabilizers and right cosets rearrange the same ensemble average inside each application of $\mathcal{L}$. Both levels are verified against unreduced calculations. The cubic test reconstructs Raman and IR spectra at 10^{-14} relative accuracy with one independent run per observable, and the measured timing improvement is substantial once the recurrence exceeds fixed checkpoint overhead.

The resulting code has one optical-definition layer, one workflow layer, and small backend adapters. Obsolete KPM code is gone and unvalidated two-phonon code cannot be invoked accidentally. This provides a clean base on which future optical vertices can be added only after their physics and symmetry are independently validated.

A Algebraic equivalence of Raman normalizations

Let a raw component be $\mathbf{v}$ with response $\mathcal{R}_{\mathbf{v}}$. Because a diagonal linear response is bilinear in its two external vertices,

$$\mathcal{R}_{s\mathbf{v}} = s^2 \mathcal{R}_{\mathbf{v}}. \quad (33)$$

For the isotropic component $s = 1/3$, so

$$45\mathcal{R}_{\mathbf{v}/3} = 45\frac{1}{9}\mathcal{R}_{\mathbf{v}} = 5\mathcal{R}_{\mathbf{v}}. \quad (34)$$

For a diagonal deviator $s = 1/\sqrt{2}$, $7s^2 = 7/2$. For an off-diagonal component $s = \sqrt{3}$, $7s^2 = 21$. This proves every row of Table 1; it also explains why saved spectra cannot be combined safely unless their component convention is known.

B Why a zero component needs no Lanczos run

If $\mathbf{v} = \mathbf{0}$, then the external coupling in Equation (2) vanishes. Bilinearity gives

$$G_{0,0}(\omega) = 0, \quad \mathcal{R}_0(\omega) = 0. \quad (35)$$

Attempting Lanczos recursion would require normalizing the zero initial vector and is undefined. Therefore “record zero, schedule nothing” is both the mathematically correct behavior and the numerically safe behavior.

C Manifest interpretation of symmetry reconstruction

For each requested component the manifest stores

$$(\text{run_id}, w_j, \chi_j, j). \quad (36)$$

A null `run_id` means the component is identically zero. Otherwise all components with the same identifier share one cached diagonal Green function. Because $G_{-v,-v} = G_{v,v}$, χ_j is not multiplied into a diagonal intensity. It remains essential metadata for any future G_{v_i,v_j} cross response, where the relative sign does matter.

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